



21CSE399T – Advanced Mobile Communications

VI Semester 2021 Regulation - Elective

Unit-3

5G Network and Standards

Content

- New Radio (NR)
- Standalone and Non-standalone mode
- Non-orthogonal Multiple Access (NOMA)
- Massive MIMO
- Beam formation
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- Flexible Frame Structure

New Radio (NR)

"New Radio" (NR) in mobile communication refers to the standard for the next generation of wireless network technology, commonly known as 5G, which provides significantly faster data speeds, lower latency, and improved connectivity compared to previous generations like 4G LTE; essentially, it's the underlying radio technology that powers 5G networks, enabling a wider range of applications from enhanced mobile broadband to massive machine-type communication (mMTC) and ultra-reliable low-latency communication (uRLLC).

New Radio (NR)

How does 5G NR Work?

- OFDM Optimization
 - 5G NR uses Orthogonal Frequency-Division Multiplexing (OFDM) for both uplink and downlink transmission. It is the same technology used in 4G/LTE networks. Thus, it also uses new features like QAM schemes (already talked about) to provide better network efficiency.
- New Higher Radio Spectrum Bands
 - 5G New Radio operates on multiple frequency bands. Consequently, these bands are categorized into two main frequency ranges:
 - Frequency Range 1 (FR1)
 - FR1 includes sub-6 GHz bands. Previous standards traditionally use these but have been extended to cover new spectrum offerings from 410 MHz to 7125 MHz. Plus, these bands offer a balance between coverage and capacity.
 - Frequency Range 2 (FR2)
 - FR2 incorporates millimeter wave (mmWave) bands, ranging from 24 GHz to 100 GHz. These higher frequency bands provide extremely high data rates and capacity. However, these bands have limited range and penetration capabilities.
- Advanced Beamforming Techniques
 - Techniques such as hybrid beamforming and coordinated beamforming are used to enhance performance further. Hybrid beamforming combines analog and digital beamforming to improve efficiency. Similarly, coordinated beamforming involves multiple base stations working together to optimize signal delivery.

New Radio (NR)

- Small Cell Coordination
 - 5G networks consist of different equipment, including small cells (like our [NanoLink](#)). Factors like power levels, frequency bands, backhaul connectivity, and coordination with macrocells should be considered during the deployments of small cells.
- Coordination Methods
 - Techniques like Coordinated Multi-Point (CoMP) and Network Function Virtualization (NFV) are used to manage small cell networks.
 - CoMP involves multiple base stations coordinating their transmissions to reduce interference and improve performance. Alternatively, NFV allows network functions to be deployed as software on standardized hardware, enhancing flexibility and scalability.
- Interference Management
 - Effective interference management is essential in small cell networks. Interference coordination and power control are used to minimize interference between small cells.

New Radio (NR)

Key points about NR:

- **Full name:** 5G New Radio
- **Function:** The air interface standard that defines how 5G devices communicate with the network
- **Benefits:** Higher data rates, lower latency, greater capacity to support a wider variety of devices and use cases
- **Key features:**
 - Advanced antenna technologies like Massive MIMO
 - Flexible spectrum utilization including high-band millimeter wave frequencies
 - Support for diverse applications like autonomous vehicles, IoT devices, and augmented reality
- **Standardization body:** 3GPP (Third Generation Partnership Project)

New Radio (NR)

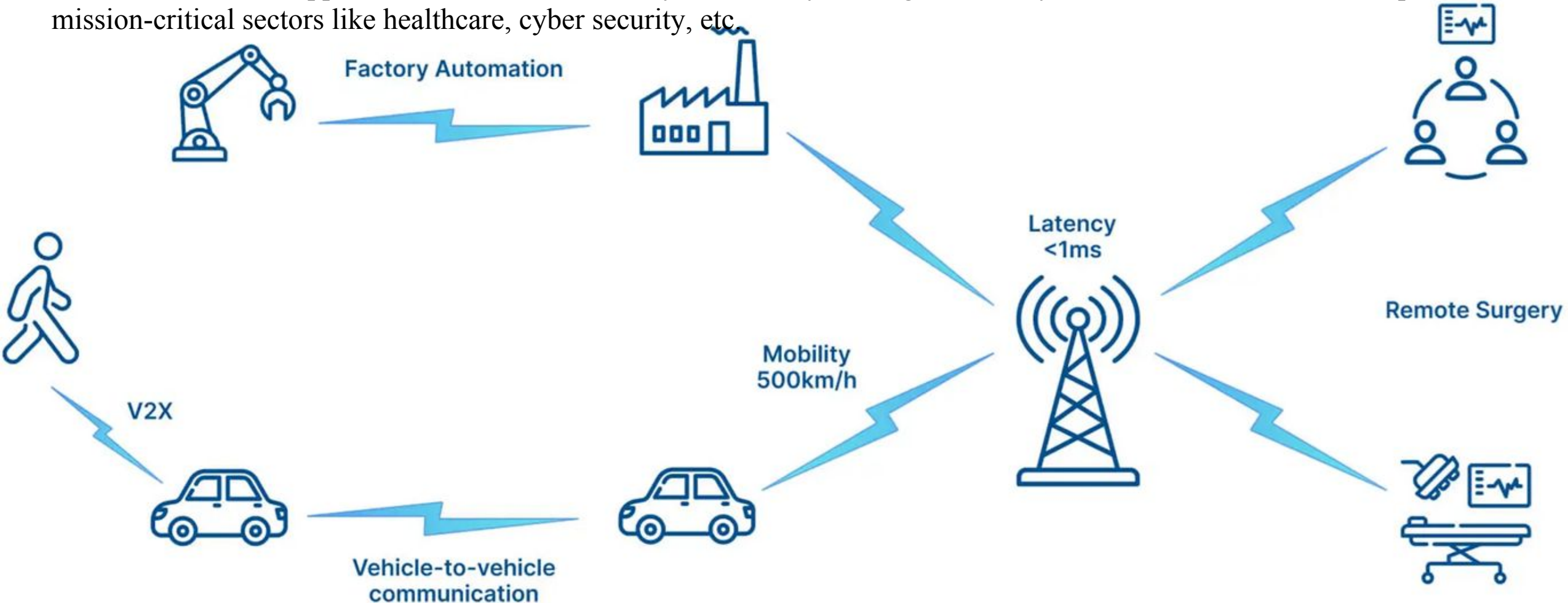
Key Features and Benefits:

- **Faster Data Speeds:**
 - 5G NR enables significantly faster data transfer rates, enabling applications like high-resolution video streaming and real-time data processing.
- **Lower Latency:**
 - It reduces the delay between data transmission and reception, crucial for applications like autonomous vehicles and remote surgery.
- **Improved Connectivity:**
 - 5G NR supports a greater number of devices and applications, including the Internet of Things (IoT).
- **Enhanced Spectrum Efficiency:**
 - It optimizes the use of radio spectrum, allowing for higher capacity and better performance.
- **Flexible Network Architecture:**
 - 5G NR is designed to be flexible and scalable, supporting various network configurations and use cases.

New Radio (NR)

Ultra-Reliable Low-Latency Communication (URLLC)

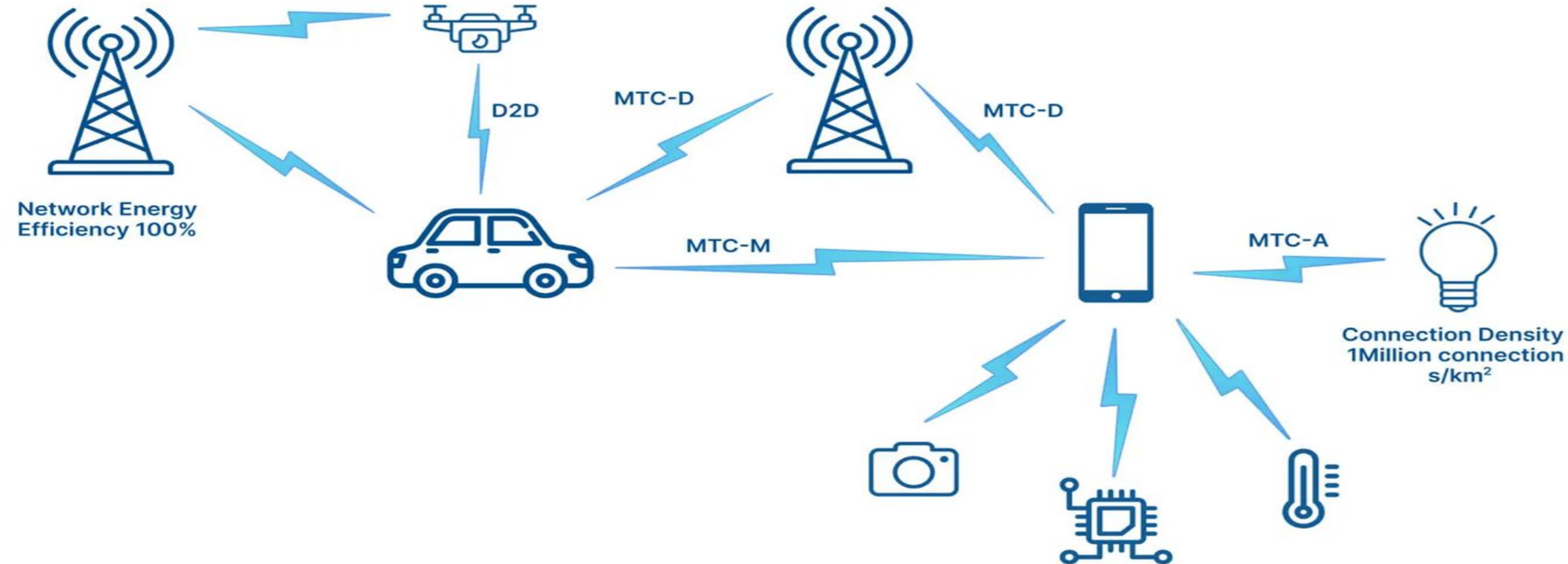
URLLC works with applications that need extremely low latency and high reliability. So, 5G NR in URLLC is important for mission-critical sectors like healthcare, cyber security, etc.



New Radio (NR)

Massive Machine-Type Communications (mMTC)

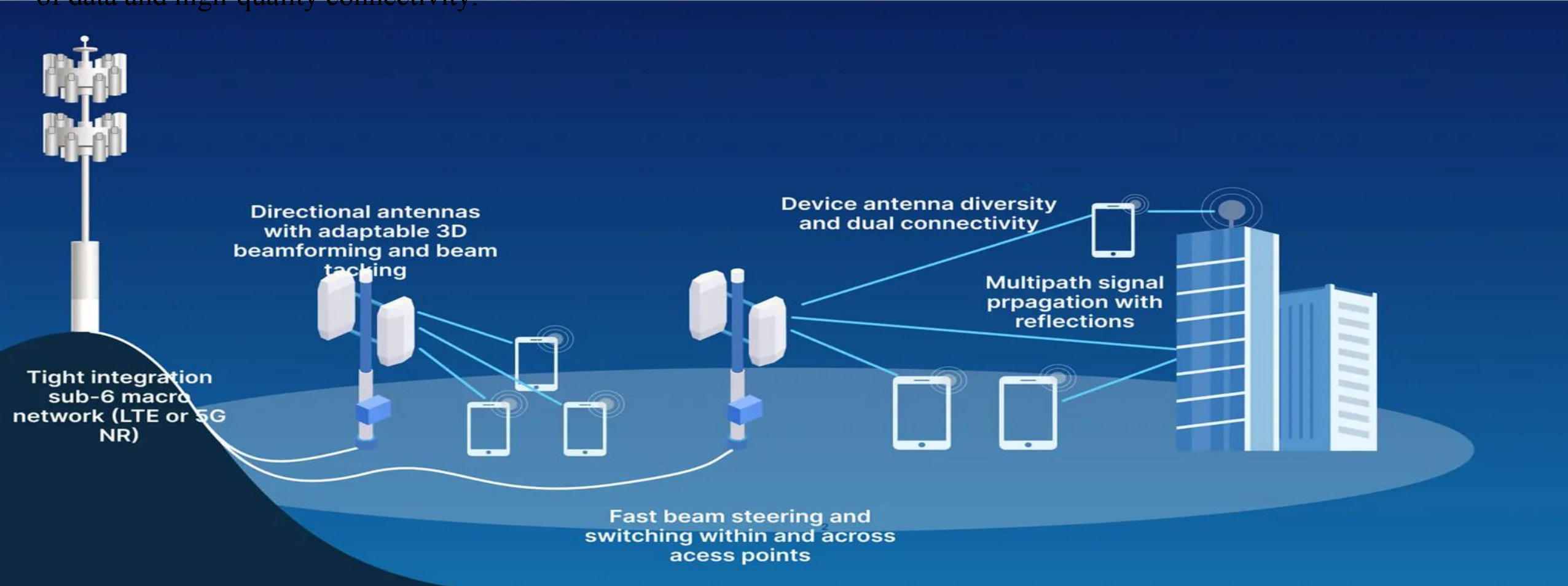
mMTC can connect many devices, which makes it ideal for Internet of Things (IoT) applications. It provides connectivity to the devices that generate small amounts of data but needs a reliable and scalable communication.



New Radio (NR)

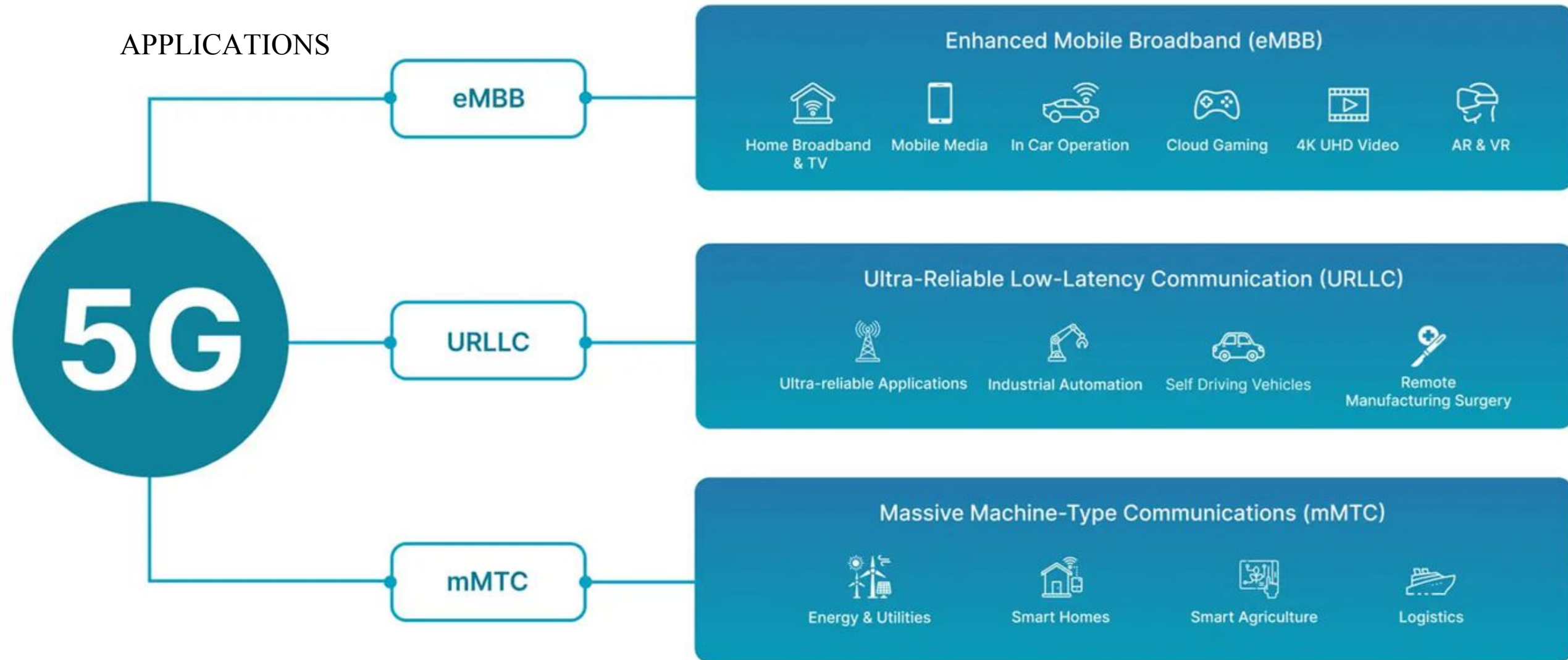
Enhanced Mobile Broadband (eMBB)

eMBB provides high data rates and improved capacity to support applications. Usually, these use cases require large amounts of data and high-quality connectivity.



New Radio (NR)

APPLICATIONS



5G NR vs. 4G LTE

New Radio (NR)

Feature	4G LTE	5G NR
Technology Standard	LTE (Long-Term Evolution)	5G New Radio (NR)
Data Rates (Average)	25 Mbps (Up to 300 Mbps peak)	100 Mbps (Up to 20 Gbps peak)
Latency	10-50 milliseconds	Around 1 millisecond
Mobility Support	Up to 350 Km/h	Over 500 Km/h
Spectrum Efficiency	Downlink (DL) - 6 bits/Hz, Uplink (UL) - 4 bits/Hz	Downlink (DL) - 30 bits/Hz, Uplink (UL) - 15 bits/Hz
User Density	Up to ~2,000 devices per square kilometer	Up to ~1,000,000 devices per square kilometer
Energy Efficiency	Moderate	Significantly improved
Frequency Bands	Primarily low-band (600 MHz - 2.5 GHz)	Low-band, Mid-band (2.5 GHz - 6 GHz), High-band (above 24 GHz)
Deployment	Existing infrastructure	More flexible deployment options: Non-Standalone (NSA) mode and Standalone (SA) mode
Signal Transmission	Basic OFDM, Lower-Order QAM	Advanced OFDM, Higher-Order QAM
Beamforming	Limited	Advanced beamforming techniques
Frequency Bands	Primarily Low-band and Mid-band	Low-band, Mid-band, High-band (mmWave)

Standalone and Non-standalone mode

In 5G mobile communication, standalone (SA) networks use a dedicated 5G core network for full 5G capabilities, while non-standalone (NSA) networks leverage existing 4G LTE infrastructure for certain functions, enabling faster deployment but with some limitations.

5G Non-Standalone (NSA):

•Architecture:

- NSA combines a new 5G radio access network (RAN) with the existing 4G LTE core network (Evolved Packet Core or EPC).

•Functionality:

- The 5G RAN handles data transmission, while the 4G EPC manages control plane functions like mobility and network management.

Standalone and Non-standalone mode

Advantages:

- **Faster Deployment:** NSA allows for quicker 5G rollouts by reusing existing 4G infrastructure.
- **Cost-Effective:** It requires less investment compared to building a completely new 5G infrastructure.

Disadvantages:

- **Limited 5G Capabilities:** NSA may not fully realize the potential of 5G features like network slicing, ultra-reliable low latency communication (URLLC), and massive machine-type communication (mMTC).
- **Performance Bottlenecks:** The reliance on the 4G core can lead to performance limitations.

Standalone and Non-standalone mode

5G Standalone (SA):

- **Architecture:**

- SA features a dedicated 5G RAN and a cloud-native 5G core network.

- **Functionality:**

- The 5G core handles both the control and data plane, enabling a more flexible and powerful network.

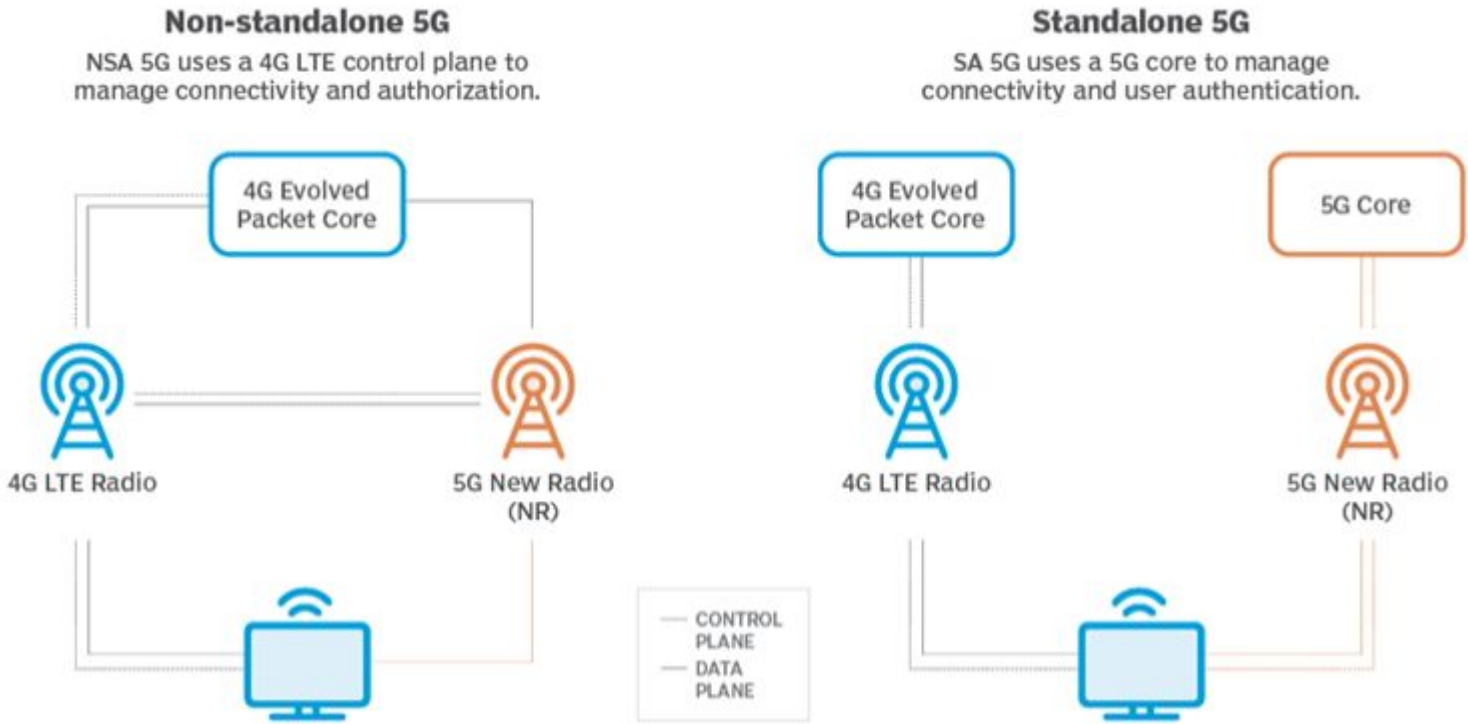
- **Advantages:**

- **Full 5G Capabilities:** SA unlocks the full potential of 5G, including network slicing, URLLC, and mMTC.
 - **Enhanced Performance:** SA offers better performance, lower latency, and higher reliability.
 - **Foundation for Future Innovations:** SA is considered the long-term solution for 5G, providing a platform for future network innovations and use cases.

- **Disadvantages:**

- **Higher Deployment Costs:** Building a new 5G core network requires significant investment.
 - **Longer Deployment Time:** SA deployments take longer than NSA deployments.

Non-standalone 5G vs. standalone 5G



Non-Standalone (NSA) 5G:

- Uses 4G LTE control plane for connectivity and authorization.
- 5G New Radio (NR) works alongside 4G LTE.
- Relies on the 4G Evolved Packet Core for network management.
- Easier to deploy as it builds on existing 4G infrastructure.

Standalone (SA) 5G:

- Uses a 5G core for full 5G connectivity and authentication.
- Eliminates dependency on 4G LTE.
- Provides better latency, network slicing, and full 5G capabilities.
- Requires new infrastructure deployment.

Feature	5G NSA	5G SA
Core Network	Existing 4G LTE core	Dedicated 5G core
Deployment	Faster, cost-effective	Slower, higher cost
Capabilities	Limited 5G features	Full 5G features
Future	Interim solution	Long-term solution, foundation for 5G advancements

Non-Orthogonal Multiple Access (NOMA)

- Non-Orthogonal Multiple Access (NOMA) is a key technology for 5G and beyond, enabling multiple users to share the same time and frequency resources, unlike traditional Orthogonal Multiple Access (OMA) which requires separate resources for each user. This allows for higher spectral efficiency and improved user capacity.
- **What is NOMA?**
 - NOMA allows multiple users to transmit data simultaneously on the same frequency and time resources, but with different power levels.
- **How it works:**
 - **Power Domain NOMA:** The base station transmits a superposition of signals from different users, each with a different power level, to the same receiver.
 - **Code Domain NOMA:** Users are distinguished by different codes or spreading sequences.

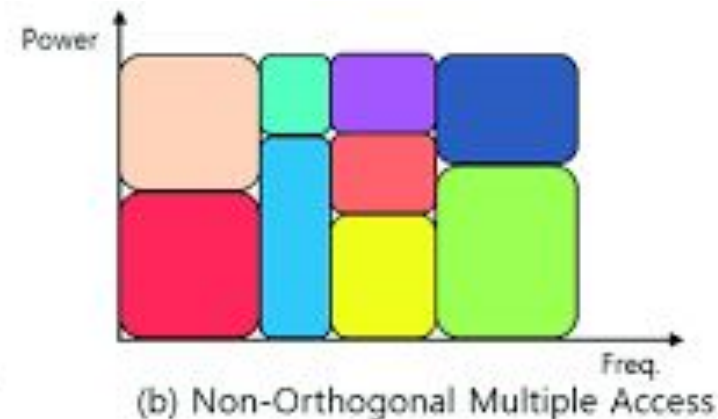
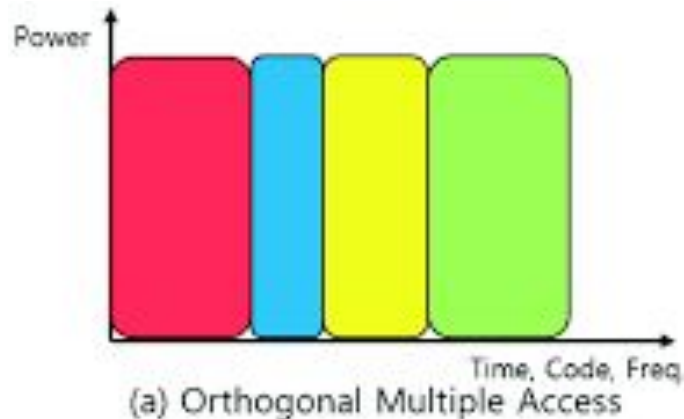
Non-Orthogonal Multiple Access (NOMA)

- **Why is it important for 5G?**
 - **Higher Spectral Efficiency:** NOMA can accommodate more users in the same bandwidth compared to OMA, leading to higher data rates and capacity.
 - **Massive Connectivity:** NOMA is well-suited for supporting a large number of devices in densely populated areas.
 - **Improved Fairness:** NOMA can be designed to provide better fairness to users with different channel conditions.
- **Comparison with OMA:**
 - **OMA (e.g., OFDMA):** Users are assigned separate time and frequency resources, leading to lower spectral efficiency.
 - **NOMA:** Users share the same resources, but the receiver uses techniques like successive interference cancellation (SIC) to separate the signals.

Non-Orthogonal Multiple Access (NOMA)

- **Challenges and Future Directions:**

- **Receiver Complexity:** NOMA receivers require more complex signal processing to separate the signals from different users.
- **Power Allocation:** Efficient power allocation is crucial for maximizing system performance.
- **Cooperative NOMA:** Combining NOMA with cooperative communication techniques can further improve performance.



Non-Orthogonal Multiple Access (NOMA)

Orthogonal Multiple Access (OMA)

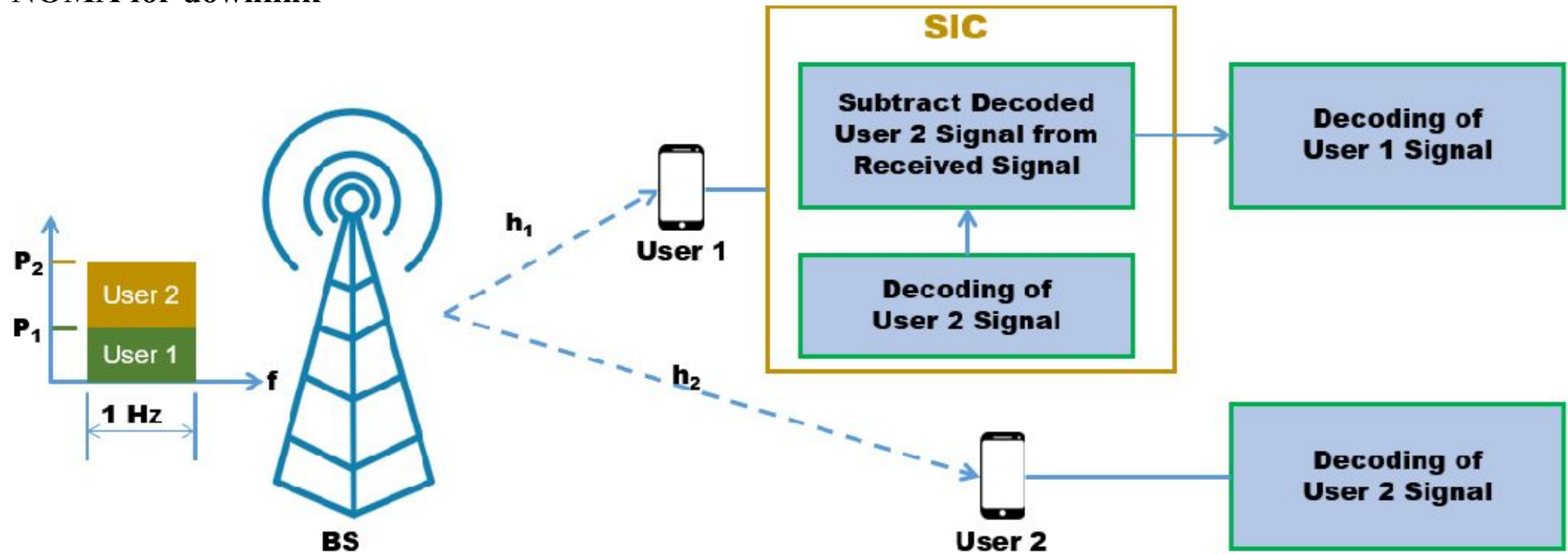
- Allocates distinct time, frequency, or code resources to users.
- Users do not interfere with each other.
- Examples: TDMA, FDMA, CDMA, and OFDMA (used in LTE).
- Limitation: Spectral efficiency is lower as resources are strictly partitioned.

Non-Orthogonal Multiple Access (NOMA)

- Allows multiple users to share the same frequency and time by using different power levels.
- More efficient spectrum usage compared to OMA.
- Key technology in 5G and beyond.
- Challenge: Requires complex Successive Interference Cancellation (SIC) to decode signals.

Non-Orthogonal Multiple Access (NOMA)

NOMA for downlink



2-user power-domain downlink NOMA.

Non-Orthogonal Multiple Access (NOMA)

1. Base Station Transmission

- The BS superimposes the signals for User 1 (near user) and User 2 (far user) in the same frequency band (1 Hz).
- **Power Allocation:**
 - User 1 (stronger channel, near BS) gets lower power (P_1).
 - User 2 (weaker channel, far from BS) gets higher power (P_2).
- The transmitted signal from the BS is:

$$s = \sqrt{P_1}x_1 + \sqrt{P_2}x_2$$

where:

- x_1, x_2 are the messages for User 1 and User 2.
- P_1, P_2 are the allocated power levels.

Non-Orthogonal Multiple Access (NOMA)

2. Reception at the Users

- Both users receive the same superimposed signal but experience different channel gains (h_1, h_2).
 - User 1 (near user, strong channel gain h_1) receives more interference from User 2's signal.
 - User 2 (far user, weak channel gain h_2) mainly decodes its own signal.

3. Successive Interference Cancellation (SIC)

- The users decode their signals using SIC, but in reverse order compared to uplink NOMA.

Step 1: User 2 Decodes its Own Signal (No SIC needed)

1. User 2 has weak channel conditions but was assigned higher power (P_2).
2. It directly decodes its own signal while treating User 1's signal as noise.

Step 2: User 1 Performs SIC

3. Since User 1 is closer to the BS, it has a stronger channel gain (h_1).
4. It first decodes User 2's signal, because User 2's signal has higher power (P_2).
5. Once decoded, User 1 subtracts User 2's signal from the received signal.
6. After cancellation, only User 1's signal remains, which is then decoded without interference.

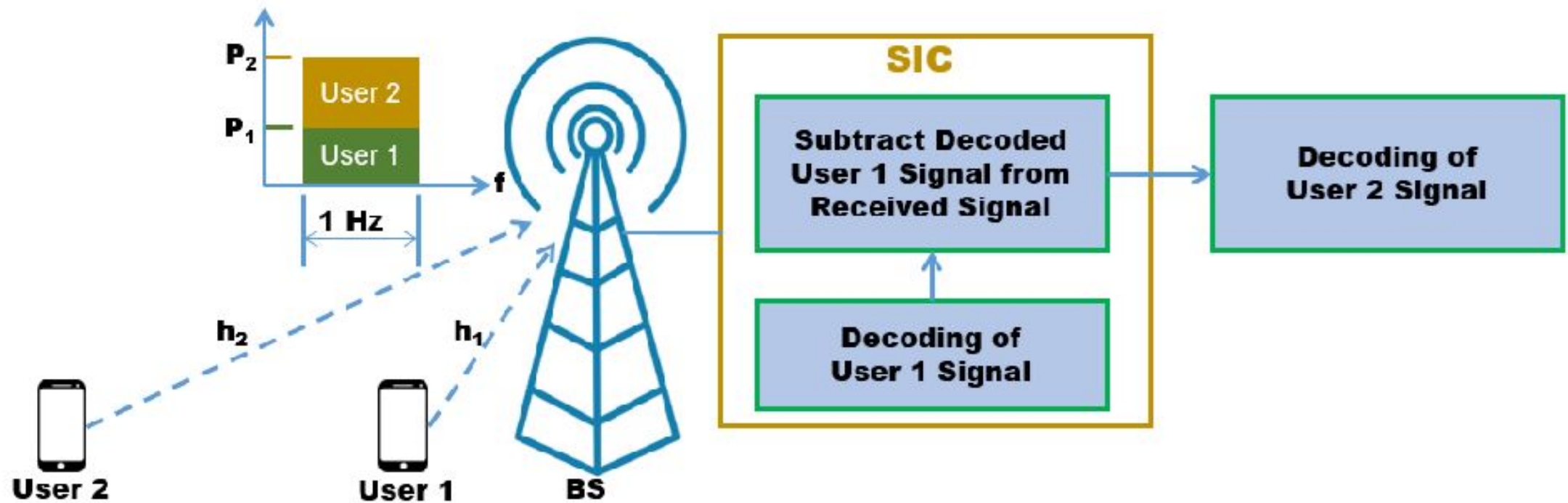
Non-Orthogonal Multiple Access (NOMA)

NOMA for downlink

- In NOMA downlink, the base station superimposes the information waveforms for its serviced users. Each user equipment (UE) employs SIC to detect their own signals.
- In the network, it is assumed that the h_1 is the closest to the base station (BS), and h_2 is the farthest.
- The challenge for BS is to decide how to allocate the power among the individual information waveforms, which is critical for SIC.
- In NOMA downlink, more power is allocated to UE located farther from the BS and the least power to the UE closest to the BS. In the network, all UEs receive the same signal that contains the information for all users.
- Each UE decodes the strongest signal first, and then subtracts the decoded signal from the received signal.
- SIC receiver iterates the subtraction until it finds its own signal. UE located close to the BS can cancel the signals of the farther UEs. Since the signal of the farthest UE contributes the most to the received signal, it will decode its own signal first.

Non-Orthogonal Multiple Access (NOMA)

NOMA for uplink



2-user power domain uplink NOMA.

Non-Orthogonal Multiple Access (NOMA)

1. **Base Station (BS):** Receives signals from multiple users.
2. **User 1 & User 2:** Two devices communicating with the base station.
3. **Power Allocation:**
 1. User 1 has power P_1 (lower power).
 2. User 2 has power P_2 (higher power).
4. **Channel Gains:**
 1. h_1 and h_2 represent the channel gains from User 1 and User 2 to the base station.

How Power Domain NOMA Works in Uplink?

- Both users transmit their signals over the same frequency (1 Hz) at different power levels.
- The base station applies Successive Interference Cancellation (SIC) for signal separation:
 - Decode the signal of User 1 first (since it has lower power).
 - Subtract User 1's signal from the received signal.
 - Decode the remaining part as User 2's signal (which had a higher power).

Why SIC?

- Since User 1's signal is received with a lower power, it is decoded first.
- Once decoded, it is removed from the received signal, leaving only User 2's signal for decoding.
- This enables efficient spectrum utilization and improves spectral efficiency.

Non-Orthogonal Multiple Access (NOMA)

1. User Transmissions

- User 1 and User 2 transmit simultaneously on the same frequency band (1 Hz).
- They have different power levels:
 - User 1 (near user) transmits with lower power (P_1) because it has a better channel condition (closer to the base station).
 - User 2 (far user) transmits with higher power (P_2) since it has a weaker channel condition (farther from the base station).

2. Signal Reception at Base Station

- The Base Station (BS) receives a superimposed signal containing both users' signals.
- This superimposed signal can be written as:

$$y = h_1x_1 + h_2x_2 + n$$

where:

- y = Received signal at the BS
- h_1, h_2 = Channel gains of User 1 and User 2, respectively
- x_1, x_2 = Transmitted signals of User 1 and User 2, respectively
- n = Noise

Non-Orthogonal Multiple Access (NOMA)

3. Successive Interference Cancellation (SIC) Process

- SIC is a **multi-stage decoding** process at the base station:

1. Step 1: Decode User 1's Signal

1. Since User 1 has the stronger channel condition (closer to BS), its signal is decoded first despite its lower power.
2. The base station extracts $x_{1 \times 1}$ from the superimposed signal.

2. Step 2: Subtract User 1's Signal from the Received Signal

1. After decoding User 1's signal, the base station subtracts it from the received signal, effectively removing its interference.

3. Step 3: Decode User 2's Signal

1. Now, the remaining part of the signal consists only of User 2's transmission, which can be decoded separately.

4. Why Use Power Domain NOMA in Uplink?

- **Efficient Spectrum Utilization:** Multiple users share the same bandwidth.
- **Improved Throughput:** Since multiple users transmit simultaneously, total network throughput increases.
- **Fairness in Resource Allocation:** By assigning different power levels, both near and far users can be accommodated efficiently.

Non-Orthogonal Multiple Access (NOMA)

- **How it works in uplink:**

- In uplink NOMA, multiple users transmit their signals simultaneously on the same frequency or time slots.

- **Power Domain Differentiation:**

- The key to NOMA is the use of power domain differentiation. Users with stronger channel conditions (closer to the base station) are assigned lower power levels, while users with weaker conditions (further away) are assigned higher power levels.

- **Successive Interference Cancellation (SIC):**

- At the receiver (base station), the signals are separated using SIC. The receiver first decodes the signal with the highest power (strongest user), then subtracts that signal from the received signal and decodes the signal with the next highest power, and so on.

Non-Orthogonal Multiple Access (NOMA)

	Advantages	Disadvantages
OMA	• Simpler receiver detection	• Lower spectral efficiency • Limited number of users • Unfairness for users
NOMA	• Higher spectral efficiency • Higher connection density • Enhanced user fairness • Lower latency • Supporting diverse QoS	• Increase complexity of receivers

Feature	Uplink NOMA	Downlink NOMA
Transmitter	Users send signals to BS	BS sends signals to users
Power Control	Near user: Low power Far user: High power	Near user: Low power Far user: High power
SIC Order	BS decodes near user first , subtracts, then decodes far user	Near user decodes far user first , subtracts, then decodes itself
Interference	BS handles superimposed signals	Users handle interference

Massive MIMO

Massive MIMO (multiple-input multiple-output) is a wireless technology that uses many antennas to improve 5G connectivity. It's used in both sub-6 GHz and mmWave bands.

How Massive MIMO is used

- Massive MIMO can be used to expand signals in specific directions.
- It can be used to create sharp beams that precisely target users.
- It can be used to focus on a limited and complex area or cover a wide sparse area.

Benefits of Massive MIMO

- Increased capacity:** Massive MIMO can increase data capacity and user throughput.
- Widen coverage:** Massive MIMO can widen cellular coverage areas.
- Beamforming:** Beamforming can direct signals to specific users and avoid interference with other users.
- Spatial multiplexing:** Spatial multiplexing can increase spectral efficiency, which leads to higher data rates.

Massive MIMO

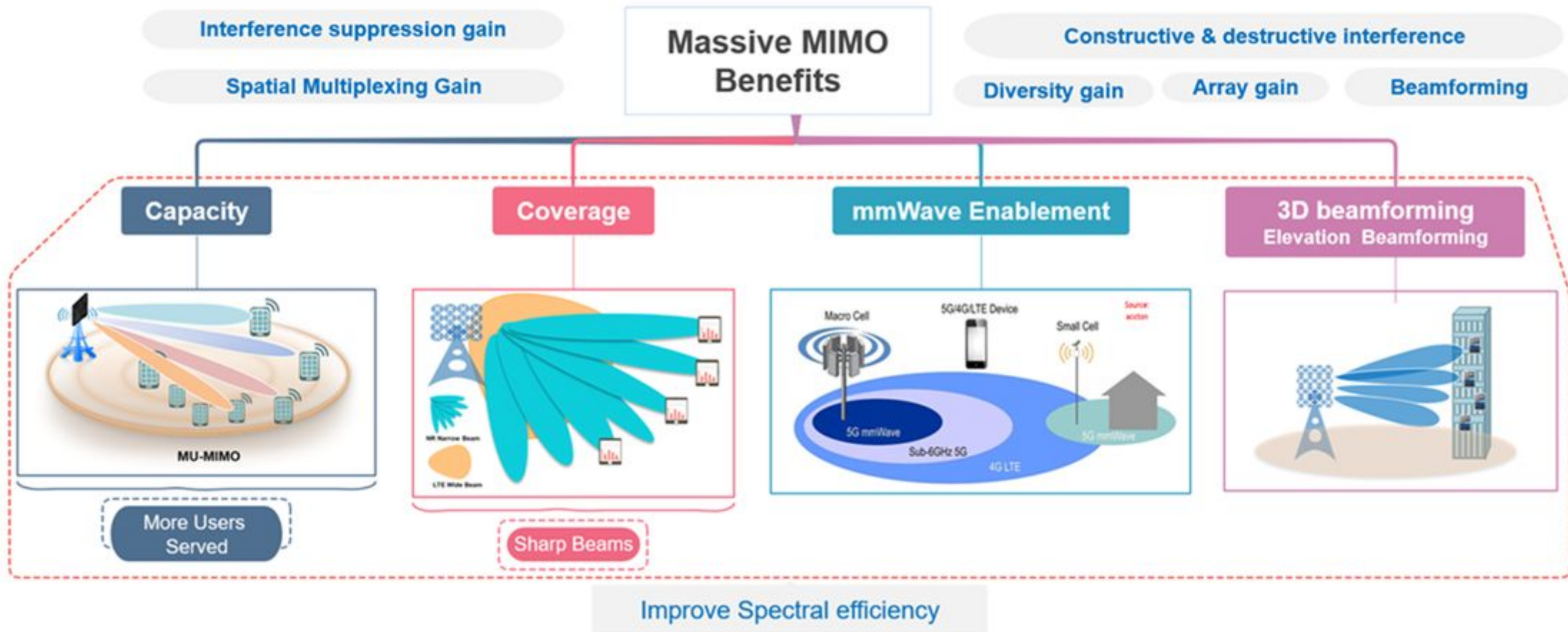
How Massive MIMO works

- 1.Massive MIMO:** Massive MIMO (Multiple Input Multiple Output) is an extension of MIMO technology, where a large number of antennas are used at the base station (gNodeB) to transmit and receive data from multiple user devices simultaneously .
- 2.More antennas:** In Massive MIMO, the base station uses a large number of antennas (e.g., 64, 128, or even more) to create multiple data streams, which enables better connectivity and higher data rates for users .
- 3.Beamforming:** Beamforming in Massive MIMO focuses the signal energy towards specific user devices, increasing signal strength and reducing interference from other users or devices. This results in better connection quality and faster data rates .
- 4.Spatial multiplexing:** Massive MIMO uses spatial multiplexing to transmit multiple data streams simultaneously, over the same frequency band. This increases the network capacity, allowing it to serve more users at the same time without congestion .
- 5.Enhanced coverage:** With Massive MIMO, the coverage area of the network is improved due to the ability of the antennas to focus signal energy in specific directions. This allows for better signal quality even at the edges of the cell, providing a more consistent user experience

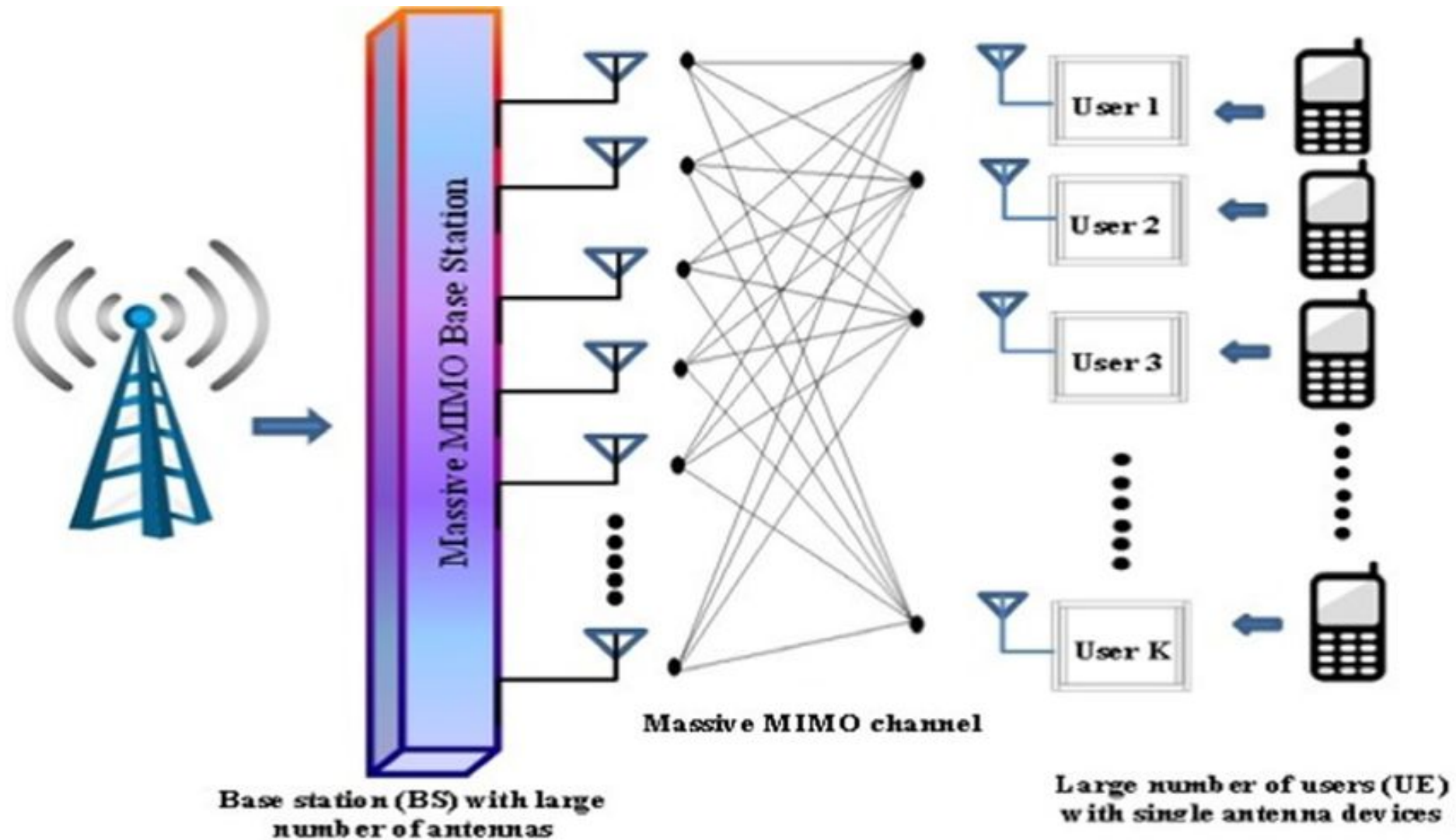
Massive MIMO

Why Massive MIMO?

- A key component of the 5G NR system is the use of Massive MIMO.
- Massive MIMO refers to an antenna array system employing a large number of antenna elements to serve multiple users simultaneously. Typically, the number of antenna elements is 128 or 256.



Massive MIMO



Massive MIMO

Challenges and Considerations:

- **Hardware Complexity:** Implementing a large number of antennas at the base station requires advanced hardware and signal processing capabilities, which can increase complexity and cost.
- **Channel Estimation:** Accurate channel state information estimation is crucial for effective precoding and beamforming. However, channel estimation becomes challenging in Massive MIMO due to increased dimensions and pilot contamination issues.
- **Backhaul Capacity:** Massive MIMO systems generate a significant amount of data traffic that needs to be transmitted between the base station and the core network. Ensuring sufficient backhaul capacity is essential to support the increased data rates.
- **Deployment Considerations:** The deployment of Massive MIMO systems requires careful planning, considering factors such as antenna spacing, interference management, and site-specific characteristics to achieve optimal performance.

Applications of Massive MIMO

- Widening cellular coverage areas
- Increasing data capacity and user throughput
- Improving network coverage, user experience, and network capability
- Enabling fast 5G rollout and expansion
- Improving coverage with intelligent automation
- Reducing inter-cell interference
- Improving capacity and load balancing

Beamforming

Beamforming is a signal processing technique in 5G that focuses transmitted signals to specific locations. It improves signal quality and reduces interference, which helps to increase network capacity.

Beamforming is used with phased array antennae systems to focus the wireless signal in a chosen direction, normally towards a specific receiving device. This results in an improved signal at the user equipment (UE), and also less interference between the signals of individual UE.

How does beamforming work?

- Uses multiple radiating elements to create a single, focused beam
- Each radiating element transmits the same signal at the same wavelength and phase
- The antenna's beam becomes narrower as it has more radiating components
- Beamforming works with MIMO (multiple input, multiple output) to improve throughput and connection density

Benefits of beamforming Improves signal-to-noise ratio (SNR), Increases signal range, Reduces interference, Supports higher-order signal modulation schemes, and Focuses transmissions toward a receiving device.

Beamforming

Three Types of Beamforming

Three main implementation approaches for beamforming are digital beamforming, analog beamforming, and hybrid beamforming. These approaches differ in the way antenna weights are modified. Antenna weights generally include amplitude and phase adjustments, with phase adjustments being more popular.

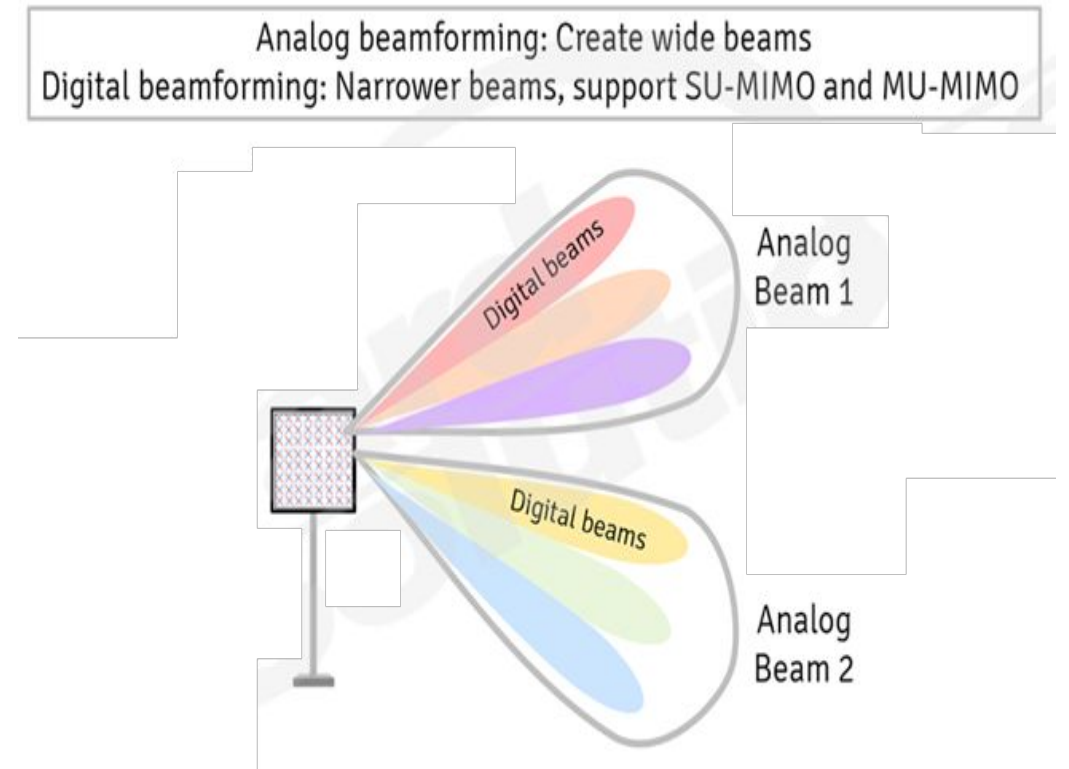
Beamforming Approaches



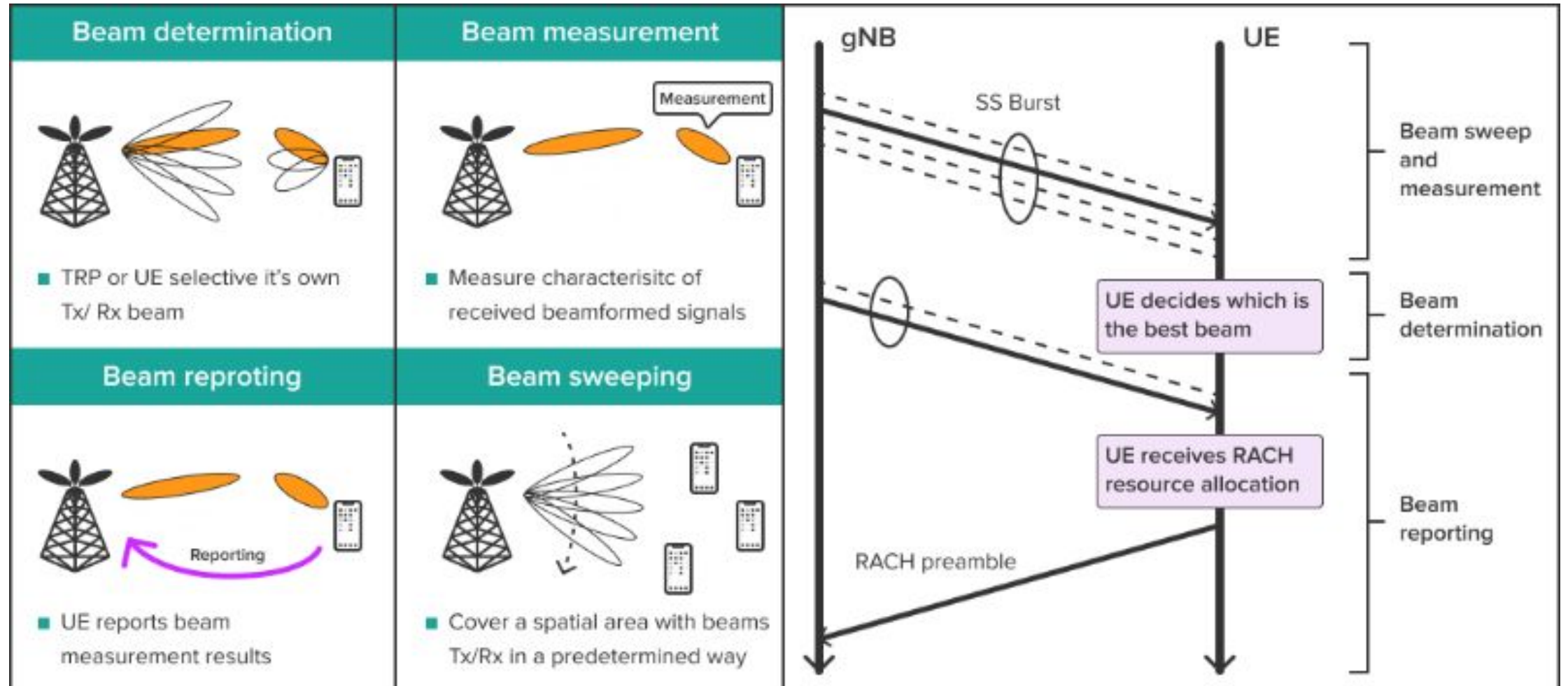
Beamforming

Analog beamforming is less expensive, but it is less flexible. In contrast, pure digital beamforming is highly flexible but expensive regarding components, processing, and costs. Thus, hybrid beamforming is the preferred choice for massive MIMO-based beamforming in 5G.

Analog beamforming creates wider beams in a cell. In contrast, digital beamforming provides the flexibility of supporting spatial multiplexing (i.e., single-user MIMO) and multi-user MIMO. Additionally, the flexibility of digital beamforming creates narrower beams (e.g., only a couple of degrees of beamwidth), simplifying the overall implementation of beamforming using massive MIMO.



Beamforming



Beamforming

Beam Management

The beam management is nothing but a procedure with a set of phases like,

- (a) **Beam sweeping** is a technique where the base station (gNB) transmits a sequence of synchronization signal blocks (SSBs) in different directions (beams) to help the user equipment (UE) find the best connection by scanning for the strongest signal
- (b) **Beam measurements** is to identify the best beam for communication by measuring the strength of received signals. .

Techniques:

- **Synchronization Signals (SSBs) in Idle Mode:** to determine the best beam for initial access.
- **CSI-RS (Channel State Information Reference Signal) in Connected Mode (Downlink):** to determine the best beam for data transmission.
- **SRS (Sounding Reference Signal) in Connected Mode (Uplink):** to determine the best beam for uplink transmission.

Metrics:

- **RSRP (Reference Signal Received Power):** UE measures the received power of the reference signals (SSBs, CSI-RS, SRS) to determine the beam strength.
- **RSRQ (Reference Signal Received Quality):** UE measures the quality of the received reference signal.
- **SINR (Signal-to-Interference-plus-Noise Ratio):** UE measures the ratio of signal power to interference and noise power.

Beam Selection Process:

- **Beam Sweeping:** The UE performs a beam sweeping process, which involves scanning a range of beams to find the one with the strongest RSRP.
- **Beam Selection:** The UE selects the beam with the highest RSRP (or other relevant metric) as the best beam for communication.
- **Beam Refinement:** After initial beam selection, further refinement can be done using CSI-RS to select a narrower beam within the initial beam.
- **Beam Switching:** The network can instruct the UE to switch to a different beam based on changing conditions or network requirements.

Beamforming

- (c) **Beam determination** involves the UE and gNB selecting the optimal beams for communication by measuring and reporting signal strength (RSRP) and channel quality, using techniques like beam sweeping and refinement, ultimately focusing the signal for improved coverage and reduced interference.
- (d) **Beam reporting** is a technique where the User Equipment (UE) informs the gNB (base station) about the best beam it has identified, facilitating efficient and robust communication, especially in millimeter-wave scenarios.
- (e) **Beam failure recovery** ensures continuous connectivity by allowing the UE to quickly re-establish a connection with the network when the signal strength of the current beam drops below a threshold, triggering a random access procedure to find a better beam.

FAPI: PHY API Specification

In 5G NR, 5G FAPI (Functional Application Platform Interface) is a suite of specifications, managed by the Small Cell Forum, that defines an interface between the MAC and PHY layers in small cells, enabling interoperability and innovation among suppliers.

Purpose:

- 5G FAPI aims to facilitate the development of small cells by providing a common API for hardware and software components, allowing different vendors to create interoperable solutions.

Key Features:

- **Interoperability:** It establishes a common standard for chipset manufacturers, component suppliers, and mobile base station integrators.
- **Flexibility:** It supports the flexible nature of 5G NR, enabling the delivery of different types of services like eMBB, URLLC, and mMTC.
- **Virtualization:** It enables the virtualization of the MAC function, as identified by the Small Cell Forum.
- **Functional Split:** FAPI defines a functional split between the MAC and PHY functions, which is important for virtualization.

FAPI: PHY API Specification

- The functional application platform interface (FAPI) is an initiative within the small cell industry released by SCF, which establishes interoperability and innovation among suppliers of platform hardware, platform software and application software. It does so by providing a common API around which suppliers may create a competitive ecosystem.
- Hence the specification helps support an innovative and competitive ecosystem for vendors of 5G small cell hardware, software and equipment. For 5G, the FAPI suite comprises five specification documents covering the following APIs:
 - 5G FAPI: PHY API – main data path (P7) and PHY mode control (P5) interface [SCF222]
 - 5G FAPI: RF and Digital Front End Control API – (P19) for Frontend Unit control [SCF223]
 - Network Monitor Mode API – (P4) for 2G/3G/4G/5G [SCF224]
 - 5G nFAPI Specification [SCF225]
 - 5G OAM Specification [SCF229]

FAPI: PHY API Specification

P5 – Control Configuration Procedures

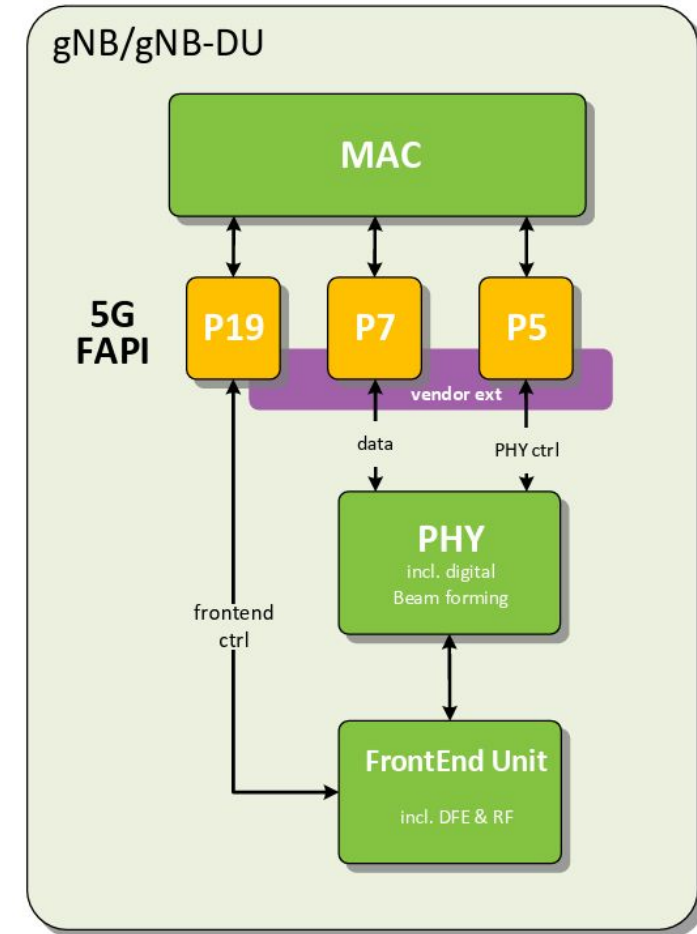
The Control configuration procedures supported by the L1 FAPI over **P5 interface** are listed below:

- Initialization
- Termination
- Restart
- Reset
- Error notification

These procedures will move the PHY layer through the IDLE, CONFIGURED and RUNNING states

P7 – Data Plane Configuration Procedures

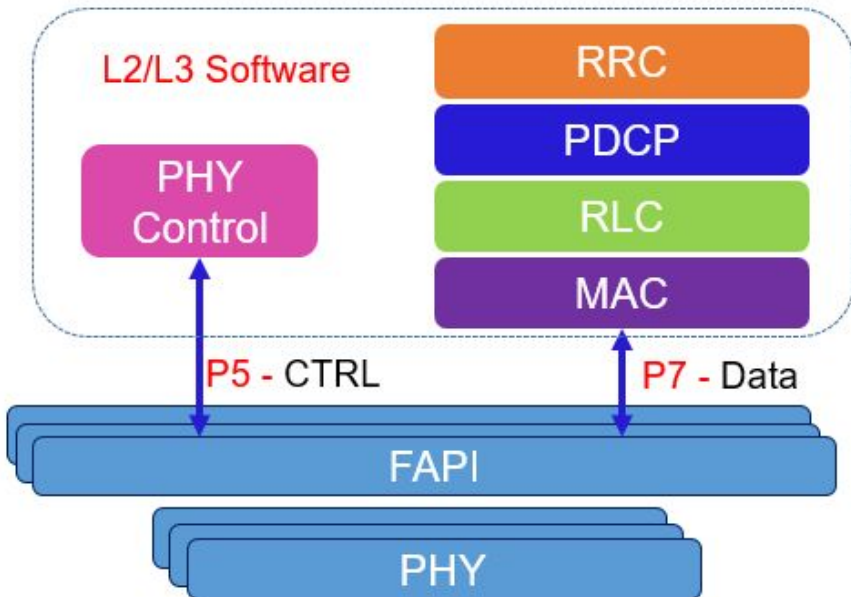
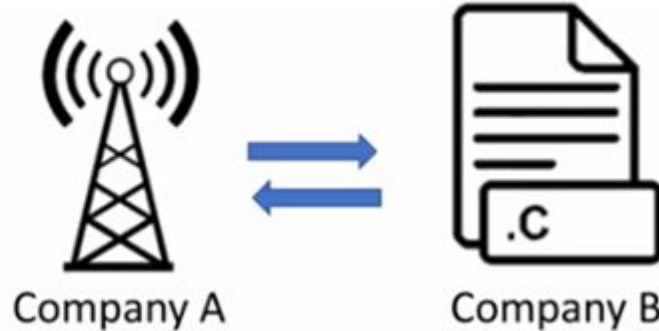
The **P7 – Data plane** configuration procedures serve two purposes for subframes procedures. Firstly, they are used to control the DL and UL frame structures. Secondly, they are used to transfer the subframe data between the **L2/L3 software** and PHY.



5G FAPI architecture

FAPI: PHY API Specification

- Both **hardware** and **software must compatible** to each other
- Ex: hardware from company A must compatible with software from company B



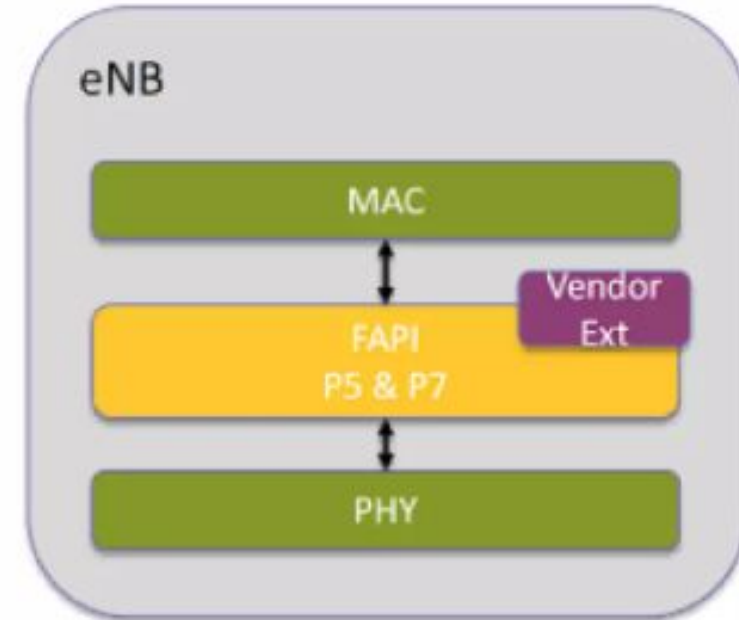
- **Small Cell Forum (SCF)** has made **universal standard** that regulates **how** software (in L2/L3) communicate with hardware (in L1)
- This standard called **FAP**I (Functional Application Platform Interface)



FAPI: PHY API Specification

What is FAPI?

- FAPI is a **standard API** that **links** between **software** in **L2/L3 layer** with **physical device/ PHY** in **L1 layer**
- FAPI consists of:
 - **P5** procedure (PHY Control) responsible to **configure physical device**
 - **P7** procedure (MAC) responsible for **data plane exchange**



FAPI: PHY API Specification

P5 Procedures Introduction

- P5 procedures **deal with PHY state**
- PHY states: **IDLE, CONFIGURED, RUNNING**
- P5 procedures:
 - **Configuration**
 - **Termination**
 - Restart
 - Reset
 - Reconfigure
 - Query
 - Notification

Example: Configuration Procedure

- Goal: **IDLE -> CONFIGURED -> RUNNING**
- Configuration procedure **has 3 stages**:
 - PARAM message exchange – **get PHY parameter**
 - CONFIG message exchange – **configure PHY**
 - START message exchange – **run PHY**
- Stages are **executed sequentially**
- PHY respond will **depend on its own state**

FAPI: PHY API Specification

P7 Procedures Introduction

- P7 procedures **responsible for data exchange**
- P7 procedures define:
 - frame, subframe, and slot number **synchronization**
 - **how** L2/L3 software instruct PHY to **transmit downlink signal**
 - **how** PHY and L2/L3 software interacts to respond **uplink signal** that comes from cell phone

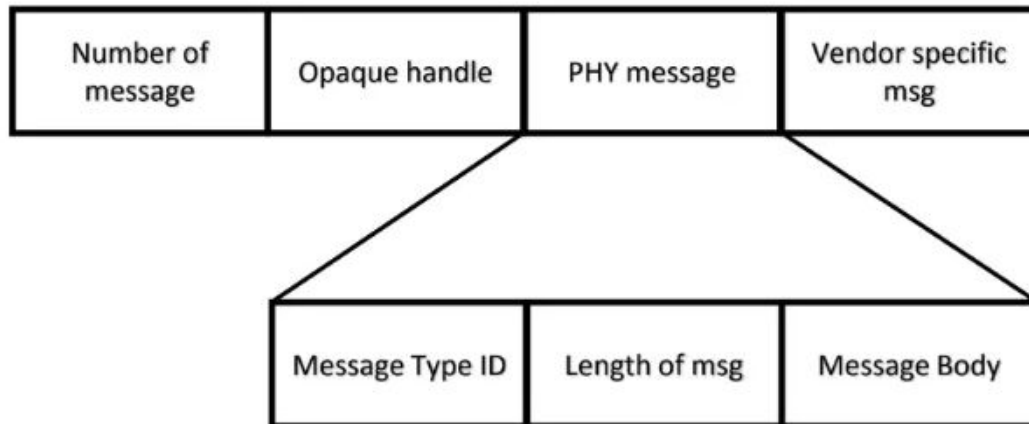
P7 Procedures

- **Slot signal**
 - **SFN/SF/SL Synchronization**
 - **Downlink Transmission Signal**
 - BCH
 - PCH
 - DL-SCH
 - Downlink reference signal
 - **Uplink Transmission Signal**
 - RACH
 - UL-SCH
 - SRS, CSI, and SR
 - Uplink reference signal
- Diagram illustrating the P7 Procedures:
- Synchronization function (Slot signal, SFN/SF/SL Synchronization)
 - How to transmit downlink signal (Downlink Transmission Signal)
 - How to respond uplink signal (Uplink Transmission Signal)

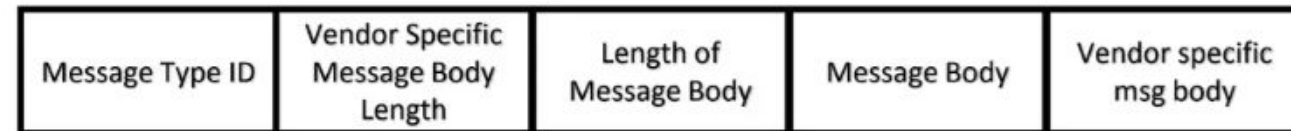
FAPI: PHY API Specification

5G FAPI vs 4G FAPI: General Message Format

5G



4G



Flexible Frame Structure

- In the context of 5G, "FAPI Flexible Frame Structure" refers to the design feature within the 5G New Radio (NR) standard that allows for dynamic allocation of time and frequency resources within a radio frame, enabling the network to adapt to diverse service needs like high-speed data, low latency communication, and massive IoT connections, essentially providing a flexible frame structure that can be customized based on application requirements, unlike previous cellular network generations with more rigid frame structures.

Flexible Frame Structure

Key points about FAPI and the Flexible Frame Structure:

FAPI (5G Femto Application Platform Interface):

- This is a set of specifications that allows different vendors to create interoperable small cell network components, which are crucial for providing dense coverage and high capacity in 5G networks.

Flexible Frame Structure:

- **Time-domain flexibility:** 5G frames can be divided into smaller time slots, allowing for dynamic resource allocation based on the needs of individual users or applications.
- **Frequency-domain flexibility:** Different subcarrier spacing options are available to optimize transmission for different frequency bands and service requirements.

Flexible Frame Structure

How it works:

Resource Grid:

- The 5G frame is divided into a grid of time and frequency resource blocks, which can be allocated dynamically to different users or services depending on their needs.

Slot-based design:

- Each subframe is further divided into slots, allowing for fine-grained control over resource allocation.

Scalable Numerology:

- The ability to adjust the subcarrier spacing based on the required bandwidth and transmission characteristics.

Flexible Frame Structure

Benefits of FAPI and Flexible Frame Structure:

Efficient resource utilization:

- The network can allocate resources optimally for different services simultaneously, maximizing throughput and minimizing latency.

Enhanced user experience:

- Different types of applications can be supported on the same network with tailored performance levels.

Network flexibility:

- Operators can easily adapt their network configuration to meet evolving service demands.

Flexible Frame Structure

The frame structure of 5G NR is in which the subframe length of 1 ms is composed of 14 OFDM symbols using 15 kHz subcarrier spacing and normal CP. The design of the 5G NR new frame structure allows uRLLC transmission over a fraction of a slot, which is referred to as mini-slot transmission. Such transmissions can also preempt an ongoing slot-based transmission to another device, allowing for prompt transmission of data requiring very low latency. Each mini-slot is also able to carry control signals/channels at the beginning and/or end OFDM symbol(s). A mini-slot is the shortest unit of resource allocation/scheduling

